

Features of Cacti

The website has a complete section dedicated to Cacti's feature set. We have reproduced the content here for your reference.

Graphs

- It allows you to create almost any imaginable RRDTool graph using all of the standard RRDTool graph types and consolidation functions.
- You can define an unlimited number of graph items for each graph, optionally using CDEFs or data sources from within Cacti.
- Automatic grouping of graph items allows quick re-sequencing of graph items.
- Auto-padding support makes sure the graph legend text lines up.

Graph display

- The tree view allows users to create 'graph hierarchies' and place graphs on the tree. This is an easy way to manage/organise a large number of graphs.
- The list view lists the title of each graph in one large list, which links the user to the actual graph.
- The preview displays all the graphs in one large list format. This is similar to the default view for all CGI scripts for RRDTool/MRTG...

Data sources

- Data sources can be created that utilise RRDTool's create and update functions. Each data source can be used to gather local or remote data to be placed on a graph.
- It supports RRD files with more than one data source and can use an RRD file stored anywhere on the local filesystem.
- Round robin archive (RRA) settings can be customised to give users the ability to gather data on a non-standard time span while storing varying amounts of data.
- It supports feeding paths to any external script/command along with any data that the user will need to 'fill in'. Cacti will then gather this data in a cron job and populate a MySQL database or round robin archives.
- Data sources can also be created, which correspond to actual data on the graph. For instance, if a user wants to graph the ping times to a host, you can create a data

source by utilising a script that pings a host and returns its value in milliseconds.

Data gathering

- It contains a 'data input' mechanism that allows users to define custom scripts that can be used to gather data. Each script can contain arguments that must be entered for each data source created, using the script (such as an IP address).
- The built-in SNMP support can use `php-snmp`, `ucd-snmp`, or `net-snmp`.
- It has the ability to retrieve data using SNMP or a script with an index. An example of this would be populating a list with IP interfaces or mounted partitions on a server. Integration with graph templates can be defined to enable one-click graph creation for hosts.
- It also provides a PHP-based poller to execute scripts, retrieve SNMP data, and update your RRD files.

Templates

- Templates allow the creation of a single graph or data source template that defines any graph or data source associated with it.
- Graph templates enable common graphs to be grouped together by templating. Every field for a normal graph can be templated or specified on a per-graph basis.
- Data source templates enable common data source types to be grouped together by templating. Every field for a normal data source can be templated or specified on a per-data source basis.
- Host templates are a group of graphs and data source templates that allow you to define common host types. On the creation of a host, it will automatically take on the properties of its template.

User management

- User-based management allows administrators to create users and assign different levels of permissions to the Cacti interface.
- You can even specify permissions per-graph for each user. This makes Cacti suitable for co location situations.
- All users can keep their own graph settings for varying viewing preferences.

Setting it up

The installation part is easy; it should be done from your distribution's software repository. If not automatically pulled in, you need to install Apache, MySQL, PHP, `php-mysql`, `net-snmp`, `php-snmp` and `rrdtool` as dependencies for Cacti. If you want to install the latest version of Cacti manually, you can find it on the downloads page of the project website at www.cacti.net/download_cacti.php

Step 1—Configure SNMP(Optional): Use the `snmpconf`

tool to create a basic set-up of SNMP for each host as follows:

```
snmpconf -g basic_setup
```

Check net-snmp.sourceforge.net/docs/man/snmpconf.html for more information on it.

If you are using SNMPv3, you will have to create a SNMP user to allow read-write access as follows: